

The Relationship Between Body Mass Index and Sleep Apnea Severity

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Project repository: <https://github.com/DeviKamakshi/bmi-sleep-apnea-regression>

Abstract

Obstructive sleep apnea is a sleep disorder in which breathing repeatedly stops or becomes reduced during sleep. This study examines the relationship between Body Mass Index (BMI) and sleep apnea severity, measured using the Apnea-Hypopnea Index (AHI), among 100 participants in the DREAMT study. Simple linear regression was used to estimate the relationship between BMI and AHI. A one-unit higher BMI was associated with an expected AHI that was approximately 1.41 events per hour higher, and the linear relationship with BMI explained 17.8% of the observed variation in AHI. However, some assumptions of the regression model were not fully satisfied. The results suggest that BMI is associated with sleep apnea severity, but BMI alone explains only part of the variation in AHI between individuals.

1 Introduction

Sleep apnea is a sleep disorder in which a person's breathing repeatedly stops or becomes reduced while sleeping. These interruptions can affect sleep quality and may also affect a person's health. One common measure used to describe the severity of sleep apnea is the Apnea-Hypopnea Index (AHI). AHI measures the number of apnea and hypopnea events that occur per hour of sleep. A higher AHI generally indicates more frequent breathing interruptions during sleep.

Sleep apnea can be associated with several factors, including age, genetics, upper airway anatomy, body weight, smoking, alcohol use, and physical activity. Among these factors, Body Mass Index (BMI) is useful to study because it is an easily measured characteristic based on a person's weight relative to height. Understanding how BMI is related to sleep

apnea severity may also have practical value. People with higher BMI who already experience symptoms such as snoring, poor sleep, or daytime tiredness may benefit from discussing sleep apnea testing with a healthcare provider. Earlier identification of sleep apnea may allow patients and healthcare providers to consider appropriate treatment, including lifestyle changes or positive airway pressure treatments such as CPAP or BiPAP when appropriate.

Previous research has studied the relationship between body weight and sleep apnea in several ways. Peppard et al. followed 690 adults over four years to determine whether changes in body weight were associated with changes in sleep-disordered breathing. They found that a 10% increase in body weight was associated with an approximately 32% increase in AHI, while weight loss was associated with decreases in AHI [1]. Newman et al. studied 2,968 adults in the Sleep Heart Health Study and examined changes in weight and sleep-disordered breathing over approximately five years. They found that weight gain was associated with worsening sleep-disordered breathing and that the relationship differed between men and women [2]. More recently, Esmaeili et al. analyzed 12,860 adults from four community-based cohorts. They found that obstructive sleep apnea was more common among people with overweight or obesity, but also found that many people with sleep apnea did not have obesity [3]. Together, these studies show that body weight is related to sleep apnea while also suggesting that BMI alone cannot completely explain differences in sleep apnea severity.

This study examines the relationship between BMI and AHI among participants in the DREAMT study. BMI is used as the explanatory variable and AHI is used as the response variable. Simple linear regression is used to estimate the average linear relationship between the two variables. The analysis found a positive association between BMI and AHI, although there was considerable variation in AHI among participants. This suggests that BMI is related to sleep apnea severity but does not explain all differences in AHI between individuals. Because BMI was observed rather than assigned as part of an experiment, the results describe an association rather than a causal relationship.

The remainder of this paper is organized as follows. Section 2 describes the DREAMT dataset and the variables used in the analysis. Section 3 explains the simple linear regression model and statistical methods. Section 4 presents the results, and Section 5 discusses the findings, limitations, and possible directions for future research.

2 Data

The data used in this study come from the DREAMT dataset available through PhysioNet [4]. The study recruited 100 participants from the Duke University Health System Sleep Disorders Lab between May and September 2022. Participants were recruited when they came

to the sleep laboratory for a scheduled overnight sleep study. Each participant underwent overnight polysomnography, a clinical sleep study that records physiological signals while a person sleeps. Each observation in the dataset represents one participant. The dataset contains participant characteristics and sleep-related measurements, and this analysis focuses specifically on BMI and AHI.

Body Mass Index (BMI) is the explanatory variable in this analysis. BMI is calculated by dividing a person’s weight in kilograms by the square of their height in meters, giving units of kilograms per square meter (kg/m^2). The BMI values were provided directly in the DREAMT dataset and were not calculated as part of this analysis. BMI ranged from 18 to 72, with a mean of 33.66 and a median of 32.

The response variable is the Apnea-Hypopnea Index (AHI), a measure of sleep apnea severity derived from the overnight polysomnography study. AHI represents the number of apnea and hypopnea events per hour of sleep. The derived AHI values were provided directly in the DREAMT dataset and were not calculated as part of this analysis. AHI ranged from 0 to 182 events per hour, with a mean of 22.13 and a median of 13.5 events per hour.

All 100 participants had BMI and AHI values available, so no observations were removed because of missing values for these two variables. The mean AHI was higher than the median because a small number of participants had particularly high AHI values. For example, one participant had an AHI of 182 events per hour. This observation was retained because there was no clear evidence that it was a data-entry error. The same participant had an obstructive apnea-hypopnea index (OAH) of 178, which was consistent with the unusually high AHI value.

Using data from a larger study that includes people from the general population could help determine whether the relationship between BMI and AHI is similar among people who are not already seeking evaluation for sleep problems.

3 Methods

Simple linear regression was used to examine the relationship between BMI and AHI. Let Y_i represent the AHI of participant i , and let X_i represent the BMI of participant i . The model is

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, 100.$$

In this model, β_0 is the intercept and represents the expected AHI when BMI is zero. The slope, β_1 , represents the expected difference in AHI associated with a one-unit difference in

BMI. The error term, ϵ_i , represents the difference between an individual participant's AHI and the mean AHI described by the linear relationship with BMI.

The analysis was conducted in several steps. First, summary statistics were calculated for BMI and AHI to understand the range and distribution of the two variables and to identify missing or unusual observations. A scatterplot of BMI and AHI was then examined to understand the direction and form of the relationship and to determine whether a linear model was reasonable.

The linearity assumption was evaluated using the scatterplot of BMI and AHI shown later in Figure 3. The plot showed an overall upward pattern, with higher BMI values generally associated with higher AHI values. However, the observations were widely scattered around the overall trend. Participants with similar BMI values sometimes had substantially different AHI values. For example, among participants with BMI values between approximately 30 and 40 kg/m², AHI values ranged from values close to zero to values above 100 events per hour. Several participants also had unusually high BMI or AHI values. Therefore, a positive linear trend was visible, although there was considerable variation around the trend and several unusual observations.

For each participant, the residual was calculated as the difference between the observed AHI and the AHI predicted by the fitted regression model,

$$e_i = Y_i - \hat{Y}_i.$$

A positive residual indicates that the participant's observed AHI was higher than predicted by the model, while a negative residual indicates that it was lower than predicted.

The simple linear regression model assumes that the errors have constant variance. This means that the variability of AHI around the fitted regression line should remain approximately the same across different BMI values.

The residuals-versus-fitted plot in Figure 1 was used to evaluate this assumption. This plot displays the residual for each participant against the AHI value predicted by the regression model. If the constant-variance assumption is reasonable, the residuals should have approximately the same vertical spread throughout the range of fitted values.

In this analysis, the residuals showed greater spread at some of the higher fitted AHI values. Since fitted AHI increases with BMI in this model, this suggests that the prediction errors tended to be more variable at higher BMI values. Therefore, the constant-variance assumption may not be fully satisfied. The residual plot also showed some departure from a completely random pattern around zero, suggesting that the linear model does not describe all features of the relationship.

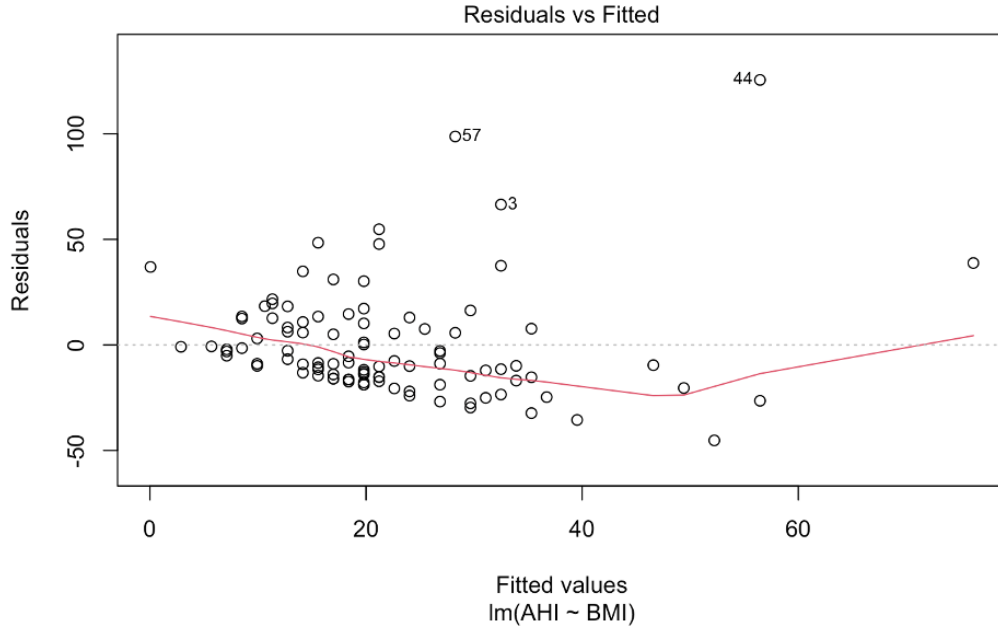


Figure 1: Residuals versus fitted AHI values for the simple linear regression model. The increasing residual spread at higher fitted values suggests that the constant-variance assumption may not be fully satisfied.

For inference about the regression slope, the errors are also assumed to be normally distributed. This assumption concerns the distribution of the errors around the regression line rather than the distribution of BMI or AHI themselves.

Normality was evaluated using the normal Q-Q plot of the residuals shown in Figure 2. A Q-Q plot compares the observed residuals with values that would be expected if the errors followed a normal distribution. If the normality assumption is reasonable, the points should fall approximately along the straight reference line.

In this analysis, the points were reasonably close to the reference line in the middle of the plot but moved noticeably above the line in the upper tail. Several large positive residuals were present, meaning that some participants had observed AHI values considerably higher than those predicted by the regression model. Therefore, the normality assumption did not appear to be fully satisfied. Because the usual t -based hypothesis test and confidence interval for the slope use the normal-error regression model, these inferential results were interpreted with caution.

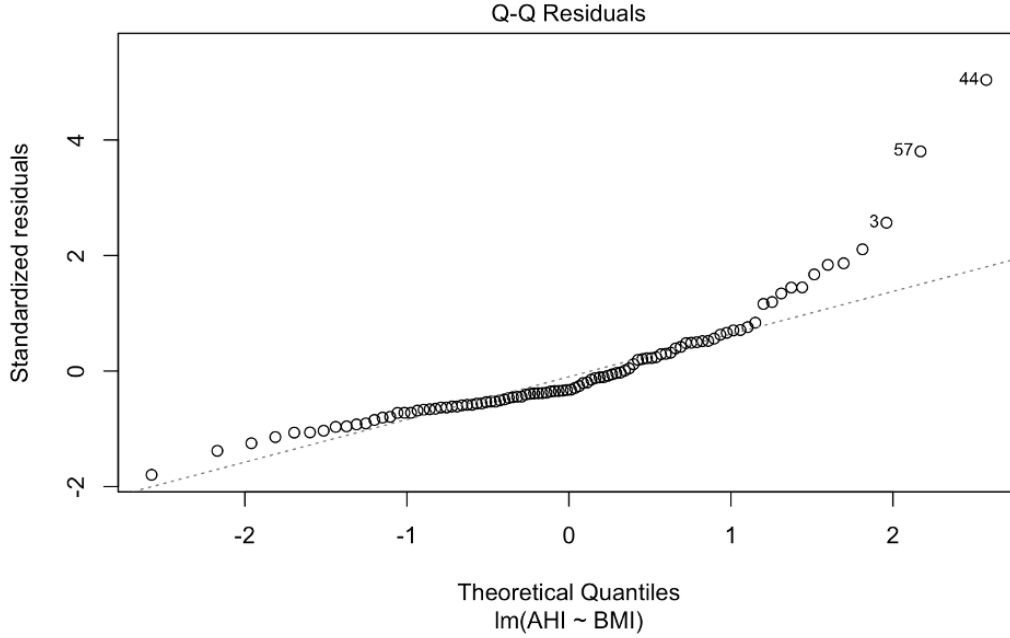


Figure 2: Normal Q-Q plot of residuals from the simple linear regression model. The noticeable departure from the reference line in the upper tail suggests that the normality assumption is not fully satisfied.

The model also assumes that errors are independent across observations. Independence was evaluated based on the structure of the data. Each observation represents a different participant, and only one BMI and one AHI value was included for each participant. Therefore, there was no obvious repeated-measures source of dependence among the observations.

After evaluating the assumptions, the parameters of the simple linear regression model were estimated using ordinary least squares (OLS), with BMI as the explanatory variable and AHI as the response variable. The model was fitted using the `lm()` function in R. The fitted regression equation and estimated slope were used to describe the direction and magnitude of the association between BMI and AHI. The coefficient of determination, R^2 , was used to describe the proportion of the observed variation in AHI explained by its linear relationship with BMI.

To determine whether the observed relationship between BMI and AHI provided evidence of a linear association in the population, a t -test was conducted for the population slope, β_1 . Different samples can produce different estimated slopes, so the test evaluates whether the estimated slope is sufficiently different from zero relative to its sampling variability. The hypotheses were

$$H_0 : \beta_1 = 0$$

and

$$H_a : \beta_1 \neq 0.$$

The null hypothesis represents no linear association between BMI and expected AHI, while the alternative hypothesis represents a linear association. A significance level of $\alpha = 0.05$ was used. A 95% confidence interval for β_1 was also calculated to describe the uncertainty in the estimated slope.

All statistical analyses were performed using R [5].

4 Results

Figure 3 shows the relationship between BMI and AHI for the 100 participants together with the fitted simple linear regression line.

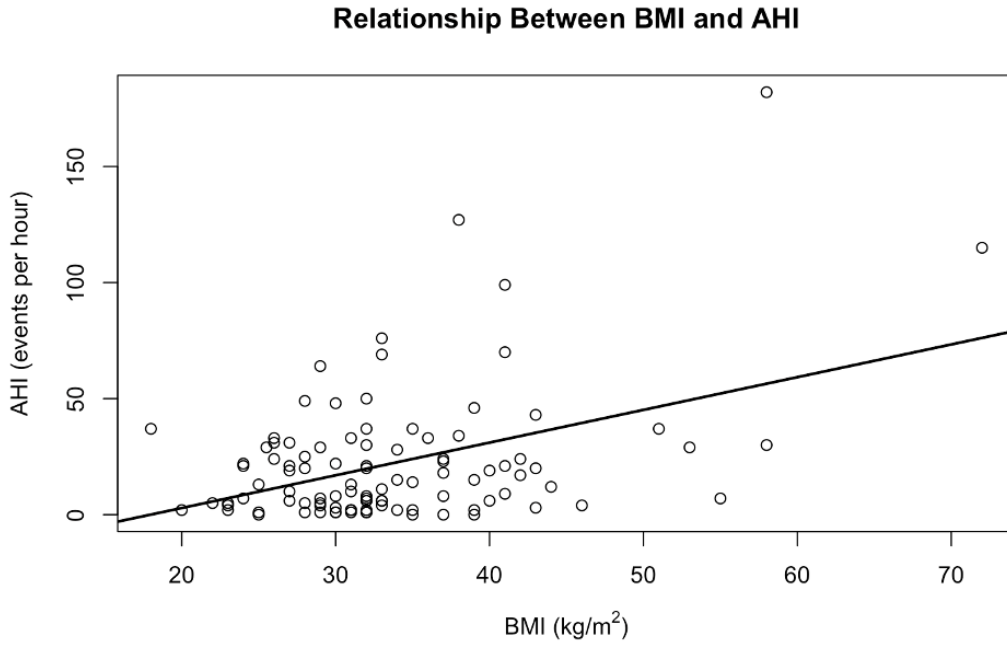


Figure 3: Relationship between BMI and AHI among the 100 DREAMT participants. The fitted regression line shows an overall positive association, although substantial variation in AHI is present among participants with similar BMI values.

The scatterplot shows an overall positive relationship, with AHI tending to increase as BMI increases. However, the observations are widely scattered around the fitted line, showing that participants with similar BMI values can have substantially different AHI values. Several participants also had unusually high AHI values compared with the majority of the sample.

The fitted regression equation was

$$\widehat{AHI} = -25.33 + 1.41(BMI).$$

The estimated slope was 1.41. This means that a one-unit higher BMI was associated with an expected AHI that was approximately 1.41 events per hour higher.

The estimated intercept was -25.33 , which represents the predicted AHI when BMI is zero. Since the observed BMI values ranged from 18 to 72 kg/m², BMI equal to zero is far outside the observed range and the intercept does not have a useful practical interpretation in this study.

The 95% confidence interval for the population slope was

$$(0.80, 2.02).$$

This interval suggests that a one-unit higher BMI is associated with an expected AHI that is approximately 0.80 to 2.02 events per hour higher. The entire interval is above zero, providing evidence of a positive linear association between BMI and AHI.

The hypothesis test for the slope produced a t -statistic of 4.61 and a p -value less than 0.001. Since the p -value was smaller than the significance level of 0.05, the null hypothesis,

$$H_0 : \beta_1 = 0,$$

was rejected. This provides statistical evidence of a linear association between BMI and AHI in the population represented by this study. However, because the constant-variance and normality assumptions were not fully satisfied, the confidence interval and hypothesis test should be interpreted with some caution.

The coefficient of determination was

$$R^2 = 0.178.$$

This means that approximately 17.8% of the observed variation in AHI among the participants was explained by its linear relationship with BMI. The remaining variation was not explained by BMI in this simple linear regression model. Therefore, although the analysis found evidence of a positive association between BMI and AHI, BMI alone explained only part of the differences in AHI observed among participants.

5 Discussion

This study examined the relationship between BMI and sleep apnea severity, measured by AHI, among 100 participants in the DREAMT study. The analysis found a positive association between BMI and AHI. A one-unit higher BMI was associated with an expected AHI that was approximately 1.41 events per hour higher. However, the R^2 value showed that the linear relationship with BMI explained only 17.8% of the observed variation in AHI. This suggests that although BMI is related to sleep apnea severity, substantial differences in AHI are not explained by BMI alone.

These findings are generally consistent with previous research. Peppard et al. found that increases in body weight were associated with increases in AHI over time [1]. Newman et al. also found that changes in weight were related to changes in sleep-disordered breathing [2]. More recent research has shown that sleep apnea occurs across different BMI levels and is not limited to people with obesity [3]. The considerable variation in AHI among participants with similar BMI values in the DREAMT data is consistent with the idea that BMI is only one of several factors related to sleep apnea severity.

There are several limitations to this analysis. First, the participants were recruited through the Duke University Health System Sleep Disorders Lab. People attending a sleep-disorders lab are likely to differ from the general adult population, especially because many may already have sleep-related symptoms or concerns. Therefore, the results may not generalize to all adults.

Second, some assumptions of the simple linear regression model were not fully satisfied. The residual variability tended to increase at higher fitted AHI values, which raised concerns about the constant-variance assumption. The residual distribution also contained several unusually large positive values, so the normality assumption was questionable. These issues mean that the usual hypothesis test and confidence interval for the slope should be interpreted cautiously.

The dataset also contained several unusual observations. In particular, one participant had a BMI of 58 kg/m² and an AHI of 182 events per hour. There was no clear reason to treat this value as a data error because the same participant had an OAH1 of 178, which was consistent with the high AHI value.

As a sensitivity check, the regression model was also fitted without this participant. The estimated slope decreased from 1.41 to 0.95 but remained positive and statistically significant ($p < 0.001$). This suggests that the unusual observation affected the estimated size of the relationship, but the positive association between BMI and AHI was not entirely dependent on that participant. The participant was therefore retained in the primary analysis.

The results should not be interpreted as evidence that higher BMI causes higher AHI. BMI was observed rather than assigned as part of a randomized experiment, and other factors may be related to both BMI and sleep apnea severity. For example, age, sex, upper airway anatomy, medical history, and other health characteristics may contribute to differences in AHI.

Future analyses could examine additional participant characteristics together with BMI to determine whether they help explain differences in AHI between individuals. It may also be useful to investigate whether the relationship between BMI and AHI differs across groups, such as younger and older adults or males and females. The scatterplot also suggested that the relationship may not be perfectly linear, so future work could consider models that allow the relationship between BMI and AHI to change across different BMI levels.

Because the DREAMT participants were recruited through a sleep-disorders lab, repeating the analysis using a larger and more representative sample would help determine whether the relationship observed in this study is similar in the broader population. Future longitudinal studies could also follow the same individuals over time to examine whether changes in BMI are accompanied by changes in AHI. This would address a different question than comparing BMI and AHI across individuals at a single point in time.

6 References

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